



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

cemc.uwaterloo.ca

**2024 Gauss Grade 7
Solutions**

1. When the four digits of 2024 are added, the result is
(A) 10 (B) 8 (C) 224 (D) 44 (E) 16

Source: 2024 Gauss Grade 7 #1

Primary Topics:

Secondary Topics:

Answer: B

Solution:

Adding the digits of 2024, we get $2 + 0 + 2 + 4 = 8$.

2. If $n = 5$, the value of $n + 2$ is
(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Source: 2024 Gauss Grade 7 #2

Primary Topics: Algebra and Equations

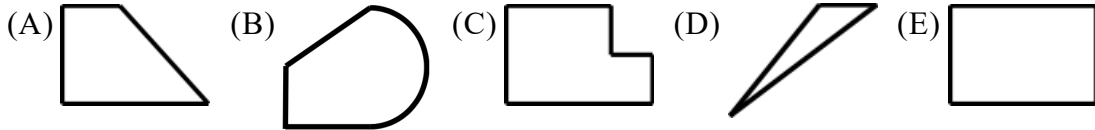
Secondary Topics: Expressions | Operations

Answer: C

Solution:

Substituting $n = 5$, we get $n + 2 = 5 + 2 = 7$.

3. Which of the following shapes has a vertical line of symmetry?



Source: 2024 Gauss Grade 7 #3

Primary Topics: Geometry and Measurement

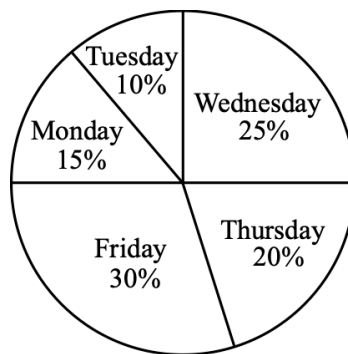
Secondary Topics: Transformations

Answer: E

Solution:

If a figure has a vertical line of symmetry, then when it is reflected in (or folded along) this vertical line, the two halves of the figure are identical to one another. Of the given shapes, such a vertical line is only possible for (E).

4. Students at Gauss Middle School were asked to choose their favourite school day. The results are shown in the circle graph.



Which day was chosen by exactly one-quarter of the students?

(A) Monday (B) Tuesday (C) Wednesday (D) Thursday (E) Friday

Source: 2024 Gauss Grade 7 #4

Primary Topics: Data Analysis

Secondary Topics: Counting | Percentages | Circles

Answer: C

Solution:

As a percent, one-quarter is equivalent to 25%. Thus, Wednesday was chosen by exactly one-quarter of the students.

5. A square with side length 5 has an area of
(A) 5 (B) 10 (C) 15 (D) 16 (E) 25

Source: 2024 Gauss Grade 7 #5

Primary Topics: Geometry and Measurement

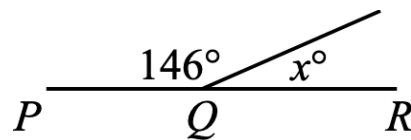
Secondary Topics: Area | Quadrilaterals | Polygons

Answer: E

Solution:

A square with side length 5 has area $5 \times 5 = 25$.

6. In the diagram, $\angle PQR$ is a straight angle.



The value of x is

- (A) 54 (B) 14 (C) 24 (D) 64 (E) 34

Source: 2024 Gauss Grade 7 #6

Primary Topics: Geometry and Measurement | Algebra and Equations

Secondary Topics: Equations Solving | Angles | Measurement

Answer: E

Solution:

Since $\angle PQR$ is a straight angle, its measure is 180° . Thus, the angles with measures 146° and x° add to 180° , and so the value of x is $180 - 146 = 34$.

7. Katie completed two laps of a track without stopping. The first lap took 3 minutes and 45 seconds, and the second lap took 4 minutes and 35 seconds. What was her total time?
- (A) 8 minutes 30 seconds
 (B) 7 minutes 50 seconds
 (C) 8 minutes 50 seconds
 (D) 8 minutes 20 seconds
 (E) 7 minutes 40 seconds

Source: 2024 Gauss Grade 7 #7

Primary Topics: Number Sense

Secondary Topics: Operations | Counting

Answer: D

Solution:

Katie's total time is equal to the sum of 3 minutes and 45 seconds and 4 minutes and 35 seconds. Adding 45 seconds and 35 seconds, we get $45 + 35 = 80$ seconds. Since there are 60 seconds in 1 minute, then 80 seconds is equal to 1 minute and 20 seconds. Thus, Katie's total time was $3 + 4 + 1 = 8$ complete minutes plus 20 seconds, for a total of 8 minutes and 20 seconds.

8. The sequence of the five symbols $\bigcirc$ $\blacktriangleleft$ $\boxtimes$ $\triangle$ $\star$ repeats to form the pattern:

$\bigcirc$ $\blacktriangleleft$ $\boxtimes$ $\triangle$ $\star$ $\bigcirc$ $\blacktriangleleft$ $\boxtimes$ $\triangle$ $\star$ $\dots$

If the pattern is continued, the 23rd symbol in the pattern is

- (A) $\bigcirc$ (B) $\blacktriangleleft$ (C) $\boxtimes$ (D) $\triangle$ (E) $\star$

Source: 2024 Gauss Grade 7 #8

Primary Topics: Algebra and Equations

Secondary Topics: Patterning/Sequences/Series | Counting

Answer: C

Solution:

Solution 1:

The sequence repeats every 5 symbols, and $23 = 4 \times 5 + 3$. Thus, the sequence of 5 symbols repeats 4 times, and 3 additional symbols follow. The 3rd symbol in the sequence is $\boxtimes$, and thus the 23rd symbol is $\boxtimes$.

Solution 2:

If the sequence of 5 symbols is repeated 4 times, then $5 \times 4 = 20$ symbols are written. Following this, 3 more symbols are needed to reach 23, and since the 3rd symbol in the sequence is $\boxtimes$, the 23rd symbol is $\boxtimes$.

9. Olivia cuts a 42 cm length of string into 2 cm pieces. Jeff cuts a 42 cm length of string into 3 cm pieces. How many more pieces of string does Olivia have than Jeff?

(A) 7 (B) 8 (C) 4 (D) 6 (E) 5

Source: 2024 Gauss Grade 7 #9

Primary Topics: Geometry and Measurement | Number Sense

Secondary Topics: Measurement | Operations

Answer: A

Solution:

Since $\frac{42 \text{ cm}}{2 \text{ cm}} = 21$, then Olivia cuts her string into 21 pieces.

Since $\frac{42 \text{ cm}}{3 \text{ cm}} = 14$, then Jeff cuts his string into 14 pieces.

Therefore, Olivia has $21 - 14 = 7$ more pieces of string than Jeff.

10. A number is randomly chosen from the list 1, 2, 3, 4, 5, 6, 7, 8, 9. The probability that the chosen number is divisible by 2, or by 3, or by both 2 and 3, is

(A) $\frac{4}{9}$ (B) $\frac{5}{9}$ (C) $\frac{6}{9}$ (D) $\frac{7}{9}$ (E) $\frac{8}{9}$

Source: 2024 Gauss Grade 7 #10

Primary Topics: Counting and Probability

Secondary Topics: Probability

Answer: C

Solution:

From the given list, the numbers that are divisible by 2 are 2, 4, 6, 8.

The numbers that are divisible by 3 are 3, 6, 9.

The numbers that are divisible by both 2 and 3 are the numbers which appear in both of the previous two lists. The only number in both lists is 6.

Thus, the numbers that are divisible by 2, or by 3, or by both 2 and 3 are 2, 3, 4, 6, 8, 9.

Since 2, 3, 4, 6, 8, 9 are 6 of the 9 numbers listed, then the probability that the chosen number is divisible by 2, or by 3, or by both 2 and 3, is $\frac{6}{9}$.

11. In the subtraction of the two-digit numbers shown, the letters P and Q each represent a single digit.

$$\begin{array}{r} 8P \\ - Q6 \\ \hline 49 \end{array}$$

The value of $P + Q$ is

- (A) 7 (B) 9 (C) 11 (D) 8 (E) 10

Source: 2024 Gauss Grade 7 #11

Primary Topics: Number Sense

Secondary Topics: Digits | Operations

Answer: D

Solution:

We rearrange the given subtraction to create the addition statement $Q6 + 49 = 8P$.

Next, we consider the units digits.

From the statement, the sum $6 + 9$ has a units digit of P which means that $P = 5$ (since $6 + 9 = 15$).

Therefore, we have $Q6 + 49 = 85$.

Rearranging this equation, we get $Q6 = 85 - 49 = 36$, and so $Q = 3$.

Therefore, $P + Q = 5 + 3 = 8$.

(We can check that $85 - 36 = 49$, as required.)

12. The length of a rectangle is twice its width. The perimeter of the rectangle is 120 cm. The width of the rectangle is

- (A) 20 cm (B) 60 cm (C) 30 cm (D) 50 cm (E) 10 cm

Source: 2024 Gauss Grade 7 #12

Primary Topics: Geometry and Measurement

Secondary Topics: Quadrilaterals | Perimeter | Expressions

Answer: A

Solution:

The perimeter of a rectangle is made up of two widths and two lengths.

Since each length is twice the width, then two lengths is equivalent to four widths, and so the perimeter is equal to $2 + 4 = 6$ times the width.

Since the perimeter of the rectangle is 120 cm, then the width of the rectangle is $\frac{120 \text{ cm}}{6} = 20 \text{ cm}$.

13. Eloise purchased a number of water hand pumps to give to a charity. The mean (average) price was \$85 per water pump. If Eloise spent a total of \$765, how many water pumps did she purchase?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 6

Source: 2024 Gauss Grade 7 #13

Primary Topics: Algebra and Equations | Number Sense

Secondary Topics: Expressions | Operations

Answer: C

Solution:

Eloise spent a total of \$765 and the mean price of each hand pump was \$85.

Thus, Eloise purchased $\frac{\$765}{\$85} = 9$ hand pumps.

14. The number 385 has three prime factors. The sum of these prime factors is

(A) 21 (B) 26 (C) 25 (D) 23 (E) 22

Source: 2024 Gauss Grade 7 #14

Primary Topics: Number Sense

Secondary Topics: Counting | Prime Numbers | Factoring

Answer: D

Solution:

Since 385 has a units digit of 5, then it is divisible by 5. Dividing, we get $\frac{385}{5} = 77$.

Since $77 = 7 \times 11$, then the three prime factors of 385 are 5, 7 and 11, and their sum is $5 + 7 + 11 = 23$.

15. A circle has radius 2. If the radius of the circle is tripled, the area of the original circle divided by the area of the new circle is

- (A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{1}{9}$ (D) $\frac{1}{2}$ (E) $\frac{1}{8}$

Source: 2024 Gauss Grade 7 #15

Primary Topics: Geometry and Measurement

Secondary Topics: Circles | Area | Fractions/Ratios

Answer: C

Solution:

A circle with radius 2 has area $\pi \times 2^2 = 4\pi$.

If the radius is tripled, then the new radius is $3 \times 2 = 6$.

A circle with radius 6 has area $\pi \times 6^2 = 36\pi$.

The area of the original circle divided by the area of the new circle is $\frac{4\pi}{36\pi} = \frac{4}{36} = \frac{1}{9}$.

16. Brett and Juanita each have a glass containing 300mL of water. Brett pours half of his water out and then Juanita pours 20% of her water into Brett's glass. What volume of water is now in Brett's glass?

- (A) 210 mL (B) 360 mL (C) 180 mL (D) 330 mL (E) 240 mL

Source: 2024 Gauss Grade 7 #16

Primary Topics: Geometry and Measurement

Secondary Topics: Percentages | Volume | Fractions/Ratios

Answer: A

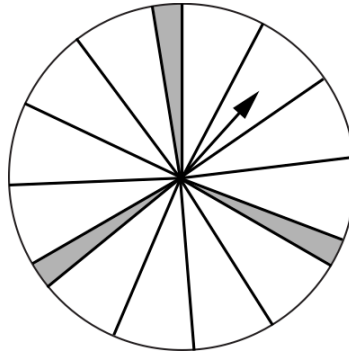
Solution:

After Brett pours half of his 300 mL of water out, he has 150 mL of water remaining in his glass.

Juanita then pours 20% of her 300 mL or $0.20 \times 300 \text{ mL} = 60 \text{ mL}$ of water into Brett's glass.

The volume of water now in Brett's glass is $150 \text{ mL} + 60 \text{ mL} = 210 \text{ mL}$.

17. A circular spinner is divided into 12 identical unshaded sections and 3 identical shaded sections, as shown.



Each unshaded section is 3 times the size of each shaded section. An arrow is attached to the centre of the spinner. The arrow is spun once. What is the probability that the arrow stops in a shaded section?

- (A) $\frac{1}{15}$ (B) $\frac{1}{5}$ (C) $\frac{1}{12}$ (D) $\frac{1}{13}$ (E) $\frac{1}{4}$

Source: 2024 Gauss Grade 7 #17

Primary Topics: Counting and Probability

Secondary Topics: Probability | Circles | Area

Answer: D

Solution:

Since each unshaded section is 3 times the size of each shaded section, then together the size of 3 shaded sections is equal to the size of 1 unshaded section.

Thus, the combined size of all sections is equal to $12 + 1 = 13$ unshaded sections.

We can now imagine the spinner as having 13 equal sized sections of which 1 is shaded.

The probability that the arrow stops in a shaded section is equal to the fraction of the spinner's area comprised of shaded sections, which is $\frac{1}{13}$.

18. The Gaussbot factory assembles robots. Each robot comes in one of three colours: red, blue, or green. Each robot also has a number stamped on its head: 1, 2, 3, or 4. The n th robot assembled is the first robot to have the same colour and the same number as a previously assembled robot. What is the greatest possible value of n ?
- (A) 11 (B) 12 (C) 13 (D) 7 (E) 8

Source: 2024 Gauss Grade 7 #18

Primary Topics:

Secondary Topics: Counting | Patterning/Sequences/Series

Answer: C

Solution:

There are 3 different colours and 4 different numbers, and therefore $3 \times 4 = 12$ different kinds of robots that may be assembled. (These are R1, R2, R3, R4, B1, B2, B3, B4, G1, G2, G3, G4, where R, B, G represent the 3 colours red, blue, green.)

Since there are 12 different kinds of robots, it is possible that the first 12 robots assembled are all different from one another.

In this case, the 13th robot assembled would be the first robot to have the same colour and the same number as a previously assembled robot, and thus the greatest possible value of n is 13.

Notes:

- i. It is possible that the first duplicate occurs earlier, but we want the latest that it can occur.
- ii. There must be a duplicate robot among the first 13 assembled, and so $n < 14$.
- iii. This solution makes use of the *Pigeonhole principle* – a concept worth further investigation.

It is possible that the first duplicate occurs earlier, but we want the latest that it can occur.

There must be a duplicate robot among the first 13 assembled, and so $n < 14$.

This solution makes use of the *Pigeonhole principle* – a concept worth further investigation.

19. Five *different* integers in a list have a median of 10 and a range of 7. What is the smallest possible integer in the list?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Source: 2024 Gauss Grade 7 #19

Primary Topics: Number Sense

Secondary Topics: Averages | Counting

Answer: B

Solution:

Consider the following list of 5 integers, ordered from smallest to largest, and having a median of 10: $a, b, 10, c, d$.

Since a is the smallest integer in the list and d is the largest, and the list has a range of 7, then d is 7 more than a .

Since a and d differ by 7, then to find the smallest possible value of a , we can find the smallest possible value of d and subtract 7.

The 5 integers in the list are different from one another, and so the smallest possible value of c is 11 (c must be greater than the median 10), and the smallest possible value of d is thus 12.

Since d is 7 more than a , then the smallest possible integer in the list is $12 - 7 = 5$.

(We note that 5, b , 10, 11, 12, where b is greater than 5 and less than 10, is such a list.)

20. A standing desk has 31 height settings, numbered from the lowest height, 1, to the highest height, 31. Since the desk is not working properly, when the up button is pressed, the desk goes up 6 settings at a time if possible, otherwise it does not move. When the down button is pressed, the desk goes down 4 settings at a time if possible, otherwise it does not move. If the desk starts at setting number 1, how many of the 31 settings will the desk be able to stop at?
- (A) 14 (B) 16 (C) 9 (D) 15 (E) 10

Source: 2024 Gauss Grade 7 #20

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Counting | Patterning/Sequences/Series | Expressions

Answer: B

Solution:

Beginning at height 1 and moving up 6 settings at a time, the desk can stop at settings 7, 13, 19, 25, and 31.

Beginning at height 31 and moving down 4 settings at a time, the desk can stop at settings 27, 23, 19, 15, 11, 7, and 3.

The desk originally begins at an odd-numbered height, 1.

Moving up 6 settings at a time, the desk can stop at only odd-numbered heights (since an even number added to an odd number is odd).

Similarly, moving down 4 settings at a time, the desk can stop at only odd-numbered heights.

Thus, it is not possible for the desk to stop at an even-numbered setting.

To this point, we have shown that the desk is able to stop at the settings

1, 3, 7, 11, 13, 15, 19, 23, 25, 27, 31,

and is not able to stop at even-numbered settings.

Next, we will show that it is possible for the desk to stop at the remaining odd-numbered settings, 5, 9, 17, 21, and 29.

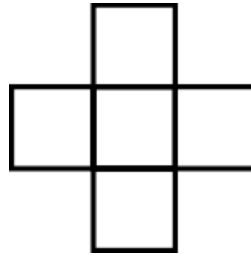
Since the desk can stop at setting 13, then it can stop at settings 9 and 5 with one and two presses of the down button, respectively.

Similarly, since the desk can stop at setting 25, then it can stop at settings 21 and 17.

Finally, since the desk can stop at setting 23, then one press of the up button will take the desk to setting 29.

The desk can stop at all odd-numbered settings from 1 to 31 inclusive, and thus is able to stop at 16 different settings.

21. Five *different* integers are selected from 1 to 6 and one integer is placed into each of the five squares shown.



The integers are placed so that the sum of the three integers in the vertical column is 7, and the sum of the three integers in the horizontal row is 11. Which integer does not appear in any square?

- (A) 3 (B) 4 (C) 2 (D) 6 (E) 5

Source: 2024 Gauss Grade 7 #21

Primary Topics: Number Sense | Other

Secondary Topics: Counting | Operations

Answer: E

Solution:

The three different integers selected from 1 to 6 and whose sum is 7 must be the integers 1, 2, 4.

Thus, the vertical column contains the integers 1, 2, 4 in some order.

(Can you see why no other combination of three of the given integers has a sum of 7?)

The three different integers selected from 1 to 6 and whose sum is 11 must be 1, 4, 6 or 2, 4, 5 or 2, 3, 6.

If the integers in the horizontal row are 1, 4, 6, then there are two integers in common with those in the vertical column, namely 1 and 4.

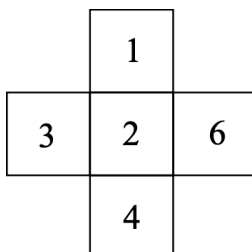
Since there have to be five different integers used in the squares, then there cannot be two integers in common between the two lists, and so 1, 4, 6 cannot appear in the horizontal row.

Similarly, 2, 4, 5 cannot appear in the horizontal row.

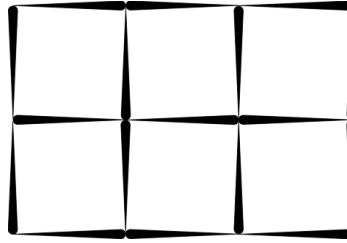
Thus, the horizontal row must contain the integers 2, 3, 6 with 2 appearing in the centre square since it is the integer in common between the two lists.

The integer not appearing in any square is 5.

The figure shows a possible arrangement of the integers.



22. In the diagram, 17 toothpicks are used to make a 2 by 3 grid of squares.



Of the toothpicks used, 10 are outer toothpicks and 7 are inner toothpicks. Suppose that toothpicks are used to make a 20 by 24 grid of squares. To the nearest percent, what percentage of toothpicks used are inner toothpicks?

- (A) 88% (B) 95% (C) 93% (D) 70% (E) 91%

Source: 2024 Gauss Grade 7 #22

Primary Topics: Number Sense | Geometry and Measurement

Secondary Topics: Counting | Perimeter | Quadrilaterals | Percentages

Answer: E

Solution:

We begin by determining the number of inner toothpicks used to make a 20 by 24 grid of squares. A grid containing 20 rows has 19 horizontal lines of inner toothpicks, each of which contains 24 toothpicks (since there are 24 columns).

Thus, the number of inner toothpicks positioned horizontally is $19 \times 24 = 456$.

A grid containing 24 columns has 23 vertical lines of inner toothpicks, each of which contains 20 toothpicks (since there are 20 rows).

Thus, the number of inner toothpicks positioned vertically is $23 \times 20 = 460$.

In total, there are $456 + 460 = 916$ inner toothpicks.

Next, we determine the total number of toothpicks used to make a 20 by 24 grid of squares.

There are 21 horizontal lines of toothpicks, each of which contains 24 toothpicks.

There are 25 vertical lines of toothpicks, each of which contains 20 toothpicks.

Thus, there are a total of $21 \times 24 + 25 \times 20 = 1004$ toothpicks used to make a 20 by 24 grid.

(Alternately, we could have determined that there are 88 outer toothpicks, and so there are $916 + 88 = 1004$ toothpicks in total.)

The percentage of inner toothpicks used is $\frac{916}{1004} \times 100\%$, which is 91% when rounded to the nearest percent.

23. A rectangular prism has integer edge lengths and has a volume of V . The six faces of the prism are painted and then the prism is cut into 1 by 1 by 1 cubes. Of these cubes, 50 cubes have no paint on them. What is the mean (average) of all possible values of V ?
- (A) 224 (B) 310 (C) 396 (D) 288 (E) 348

Source: 2024 Gauss Grade 7 #23

Primary Topics: Geometry and Measurement | Number Sense

Secondary Topics: Prisms | Volume | Surface Area | Counting | Averages

Answer: B

Solution:

All six faces of the prism are painted which means that the 1 by 1 by 1 cubes in the interior of the prism are the only cubes that have no paint on them.

Each of the three dimensions of the prism (length, width, height) must be at least 3, otherwise there are no 1 by 1 by 1 cubes without paint on them.

The set of interior 1 by 1 by 1 cubes must also be in the shape of a rectangular prism. (You should confirm each of these last two sentences for yourself before reading on.)

There are 50 interior 1 by 1 by 1 cubes, and so the volume of the interior prism is 50.

Thus, we are looking for three positive integers, representing the length, width and height of the interior prism, whose product is 50.

We may use the positive divisors of 50 (1, 2, 5, 10, 25, 50) to help identify the four possibilities: $1 \times 1 \times 50$, $1 \times 2 \times 25$, $1 \times 5 \times 10$, and $2 \times 5 \times 5$.

These are the only ways to express 50 as the product of three positive integers.

Next, we determine the dimensions of the original prisms given each set of dimensions for the interior prisms.

Consider the interior prism with dimensions $1 \times 1 \times 50$.

Recall that this is the prism that remains after all exterior (painted) cubes are removed.

That is, 1 by 1 by 1 cubes have been removed from the top and bottom of the $1 \times 1 \times 50$ interior prism, from the left and right sides, as well as from the two ends (the front and back).

This means that the dimensions of the original prism are each 2 greater than the dimensions of the interior prism. (You should try to visualize this.)

We complete the following table to determine the dimensions and the volume of the original prism in each case.

Interior prism dimensions	Original prism dimensions	Volume of original prism
$1 \times 1 \times 50$	$3 \times 3 \times 52$	$V = 3 \times 3 \times 52 = 468$
$1 \times 2 \times 25$	$3 \times 4 \times 27$	$V = 3 \times 4 \times 27 = 324$
$1 \times 5 \times 10$	$3 \times 7 \times 12$	$V = 3 \times 7 \times 12 = 252$
$2 \times 5 \times 5$	$4 \times 7 \times 7$	$V = 4 \times 7 \times 7 = 196$

Therefore, the mean of all possible values of V is $\frac{468 + 324 + 252 + 196}{4} = \frac{1240}{4} = 310$.

24. A three-digit integer is an integer from 100 to 999, inclusive. A three-digit integer is called *Tiny* if no rearrangement of its digits gives a three-digit integer that is smaller. For example, 138, 207 and 566 are Tiny, but 452, 360 and 727 are not. How many three-digit integers are Tiny?
- (A) 255 (B) 201 (C) 212 (D) 234 (E) 219

Source: 2024 Gauss Grade 7 #24

Primary Topics: Number Sense

Secondary Topics: Digits | Counting

Answer: E

Solution:

Each Tiny three-digit integer belongs to exactly one of the following three cases.

Case 1: The units digit is 0

If the units digit of a Tiny integer is 0, then the tens digit must also be 0, otherwise, the units digit and tens digit can be switched to give a smaller integer.

In this case, there are no restrictions on the hundreds digit and thus there are 9 such Tiny integers. These are: 100, 200, 300, 400, 500, 600, 700, 800, 900.

Case 2: The units digit is not 0, but the tens digit is 0

If the hundreds digit is x and the units digit is z , then the integers in this case are of the form $x0z$, where $z \neq 0$. (If x is greater than z , then switching x and z creates a smaller integer.)

Integers of this form are Tiny exactly when x is greater than or equal to 1, and x is less than or equal to z . If $x = 1$, then z can be equal to any integer from 1 to 9 inclusive, and so there are 9 such Tiny integers. These are: 101, 102, 103, $\dots$, 108, 109.

If $x = 2$, then z can be equal to any integer from 2 to 9 inclusive, and so there are 8 such Tiny integers. These are: 202, 203, 204, $\dots$, 208, 209.

Continuing in this way, there are 7 Tiny integers when $x = 3$, 6 when $x = 4$, 5 when $x = 5$, 4 when $x = 6$, 3 when $x = 7$, 2 when $x = 8$, and finally 1 when $x = 9$.

In this case, there are $9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45$ Tiny integers.

Case 3: The units digit and the tens digit are both not 0

If the hundreds digit is x (where x is greater than or equal to 1), the tens digit is y , and the units digit is z , then the integers in this case are of the form xyz . Integers of this form are Tiny exactly when x is less than or equal to y , and y is less than or equal to z .

For $x = 1$, we count the number of such Tiny integers in the table that follows.

Value of x Value of y Possible values of z Number of Tiny integers

$x = 1$	$y = 1$	$z = 1, 2, 3, 4, \dots, 9$	9
$x = 1$	$y = 2$	$z = 2, 3, 4, \dots, 9$	8
$x = 1$	$y = 3$	$z = 3, 4, \dots, 9$	7
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x = 1$	$y = 8$	$z = 8, 9$	2
$x = 1$	$y = 9$	$z = 9$	1

When $x = 1$, there are $9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45$ Tiny integers in this case.

For $x = 2$, we may similarly count the number of Tiny integers.

Value of x Value of y Possible values of z Number of Tiny integers

$x = 2$	$y = 2$	$z = 2, 3, 4, \dots, 9$	8
$x = 2$	$y = 3$	$z = 3, 4, \dots, 9$	7
$x = 2$	$y = 4$	$z = 4, \dots, 9$	6
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x = 2$	$y = 8$	$z = 8, 9$	2
$x = 2$	$y = 9$	$z = 9$	1

When $x = 2$, there are $8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 36$ Tiny integers in this case.

Notice that for each increase in the value of x by 1, the smallest possible value of y increases by 1 (to match the value of x), and so the smallest possible value of z also increases by 1 (to match the value of y).

This means that when $x = 3$, for example, the number of Tiny integers in the first row of the corresponding table is 1 less than the first row of the table for $x = 2$, and thus is 7.

That is, when $x = 3$, there are $7 + 6 + 5 + 4 + 3 + 2 + 1 = 28$ Tiny integers, and when $x = 4$, there are $6 + 5 + 4 + 3 + 2 + 1 = 21$ Tiny integers.

Continuing in this way, we summarize the count of Tiny integers for Case 3.

Value of x	Number of Tiny integers
$x = 1$	$9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45$
$x = 2$	$8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 36$
$x = 3$	$7 + 6 + 5 + 4 + 3 + 2 + 1 = 28$
$x = 4$	$6 + 5 + 4 + 3 + 2 + 1 = 21$
$x = 5$	$5 + 4 + 3 + 2 + 1 = 15$
$x = 6$	$4 + 3 + 2 + 1 = 10$
$x = 7$	$3 + 2 + 1 = 6$
$x = 8$	$2 + 1 = 3$
$x = 9$	1

The number of Tiny three-digit integers in this case is

$$45 + 36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 165$$

and so the total number of Tiny three-digit integers is $9 + 45 + 165 = 219$.

25. Suppose that

$$w, x, y, z, (x + y), (x + z), (234 + z), \text{ and } (234 - z)$$

are 8 *different* prime numbers. If $w + x + y = 234$, and each of y and z is less than 50, the value of $w - y$ is

- (A) 226 (B) 150 (C) 210 (D) 174 (E) 222

Source: 2024 Gauss Grade 7 #25

Primary Topics: Number Sense

Secondary Topics: Prime Numbers | Equations Solving | Expressions

Answer: B

Solution:

We begin by recognizing that 2 is the only even prime number.

If x , y and z are each odd prime numbers, then both $x + y$ and $x + z$ are even prime numbers (since the sum of two odd numbers is even).

However, both $x + y$ and $x + z$ are each at least $2 + 3 = 5$, and therefore each must be an odd prime number.

This tells us that x, y, z cannot all be odd prime numbers, and so exactly one of them is equal to 2 (since they are all different from one another and 2 is the only even prime number).

If $y = 2$, then each of x and z is odd and so $x + z$ is even, which is not possible.

Similarly, if $z = 2$, then each of x and y is odd and so $x + y$ is even, which is not possible, and so we conclude that $x = 2$.

Substituting $x = 2$, the list of 8 different prime numbers becomes:

$$w, 2, y, z, 2 + y, 2 + z, 234 + z, 234 - z,$$

and we note that $w + x + y = 234$ becomes $w + y = 232$.

Since z and $2 + z$ are prime numbers that differ by 2, next we consider the consecutive odd prime numbers with z less than 50.

These are: 3 and 5, 5 and 7, 11 and 13, 17 and 19, 29 and 31, 41 and 43.

So then z is equal to one of 3, 5, 11, 17, 29, or 41.

If $z = 3$, then $234 - z = 231$ which is divisible by 3 and thus not a prime number.

If $z = 11$, then $234 + z = 245$ which is divisible by 5 and thus not a prime number.

If $z = 17$, then $234 - z = 217$ which is divisible by 7 and thus not a prime number.

If $z = 29$, then $234 - z = 205$ which is divisible by 5 and thus not a prime number.

If $z = 41$, then $234 + z = 275$ which is divisible by 5 and thus not a prime number.

Finally, if $z = 5$, then $234 - z = 229$ and $234 + z = 239$, and both of these are prime numbers.

Alternately, we may have noted that if z has units digit 1, then $234 + z$ has units digit 5, and if z has units digit 9, then $234 - z$ also has units digit 5, and so each is divisible by 5, which is not possible since each is a prime number. We could have then removed $z = 11, 29, 41$ as possibilities and considered only $z = 3, 5, 17$ as we did above.

The table below summarizes what we know about the 8 different prime numbers to this point.

w	x	y	z	$x + y$	$x + z$	$234 + z$	$234 - z$
2	5	$2 + y$	7	239	229		

As shown previously, since y and $2 + y$ are consecutive odd prime numbers (with y less than 50), then y is equal to one of 3, 11, 17, 29, or 41 (recall that $z = 5$ and the 8 numbers must all be different).

Since $w + y = 232$, then $w = 232 - y$.

For which value(s) of y is $w = 232 - y$ a prime number different from those already in our list?

If $y = 3$, then $w = 229$ which is not possible since $234 - z = 229$.

If $y = 11$, then $w = 221$ which is divisible by 13 and therefore not a prime number.

If $y = 17$, then $w = 215$ which is divisible by 5 and therefore not a prime number.

If $y = 29$, then $w = 203$ which is divisible by 7 and therefore not a prime number.

Finally, if $y = 41$, then $w = 191$ which is a prime number.

The final list of 8 different prime numbers is shown below.

w	x	y	z	$x + y$	$x + z$	$234 + z$	$234 - z$
191	2	41	5	43	7	239	229

The value of $w - y$ is $191 - 41 = 150$.

